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Research Article

Defect Engineering in Atomic-Layered Graphitic Carbon Nitride for Greatly Extended Visible-Light Photocatalytic Hydrogen Evolution

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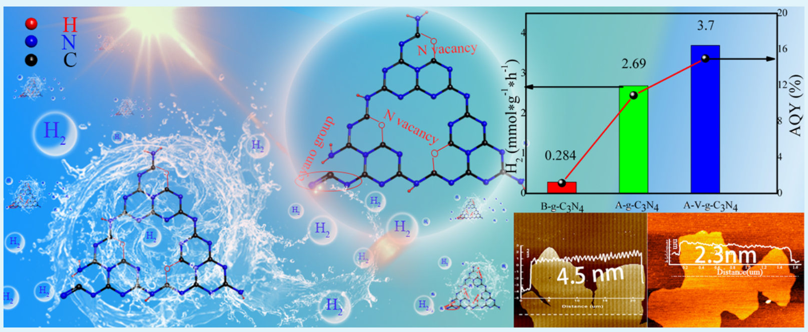

ABSTRACT: Defect modulation usually has a great influence on the electronic structures and activities of photocatalysts. Here, atomically layered g-C_3N_4 modified via defect engineering with nitrogen vacancy and cyanogen groups is obtained through two facile steps of thermal treatment (denoted as A-V-g-C_3N_4). Detailed analysis reveals that the atomic-layered graphitic carbon nitride (2.3 nm) with defect engineering modifying provides more active sites and decreases the electron/hole transferring distances. More importantly, the defects that contain nitrogen vacancies and cyanogen groups extend the responsive wavelength to 650 nm, which effectively suppresses the quantum size effect of atomic-layered g-C_3N_4. Therefore, the as-obtained A-V-g-C_3N_4 exhibited a photocatalytic H_2 evolution rate and apparent quantum yield of 3.7 mmol·g^{-1}·h^{-1} and 14.98% (λ > 420 nm), respectively. This work is expected to provide guidance for the rational design of atomic-layered g-C_3N_4.

KEYWORDS: *g-C_3N_4, defects engineering, atomic layered g-C_3N_4, quantum size effect, H_2 evolution*

1. INTRODUCTION

Semiconductor-based photocatalysis has great potential for solving energy and environmental issues by photocatalytic water splitting and organic pollutant degradation. Among many semiconductor materials, graphitic carbon nitride (g-C_3N_4) has become one of the most potential photocatalysts because of excellent photochemical stability, low cost, nontoxicity, and suitable energy band positions.[1,2] However, the photocatalytic activity on pristine materials is poor because of its inherent drawbacks, such as lower specific surface areas and poor separation efficiency of photogenerated electron/hole pairs.[3−5]

According to the photocatalytic mechanism, increasing surface active sites and enhancing the separation efficiency of e^-/h^+ pairs play a crucial role in the photocatalytic reaction.[6−10] In previous studies, many methods have been used to increase surface active sites and enhance the separation efficiency of e^-/h^+ pairs of pristine g-C_3N_4, such as coupling with other semiconductors to create heterojunctions[11−13] and design nanostructures[14,15] and so forth. Among these methods, decreasing the thickness of g-C_3N_4 nanosheets is an effective way to enhance both carrier separation efficiency and surface-active sites. It is facile to realize ultrathin g-C_3N_4 through a series of effective exfoliation means. For example, mechanical, liquid, and thermal exfoliation methods have been developed.[16−19] Their studies have shown that when the thickness of g-C_3N_4 nanosheets was decreased to the atomic-layered thickness, the migration distance of photogenerated electrons/holes was shortened, and abundant active sites were exposed. Therefore, atomic-layered g-C_3N_4 exhibit excellent photocatalytic activity. However, one open problem associated with atomically thin g-C_3N_4 is their blueshift of photo-absorption as a result of the strong quantum size effect. Therefore, atomically thin g-C_3N_4 usually has a poor capability to capture the wide-spectrum visible light and thus greatly decreased visible-light photocatalytic activity.

Based on the above analysis, to promote possible applications of atomic-layered g-C_3N_4 in solar light utilization,

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it is urgent to modify atomic-layered g-C_3N_4 with extended visible light absorption and increased visible light activities. Defect engineering has been widely investigated in photocatalysis to increase visible light absorption.[20,21] Attributing to the structure with g-C_3N_4 features about multiple coordination N atom environment, it is facile to realize defect control in g-C_3N_4.[22] For example, it was reported that thermal polymerization at high temperature,[22−24] hydrogen reduction,[25] and alkali-assisted synthesis[26,27] methods can increase light absorption under the region between 450 and 800 nm of g-C_3N_4. However, lower surface reactive sites of defect engineering modifying pristine g-C_3N_4 are obtained using these methods, which inhibited the photocatalytic activity. Thus, it is urgent to develop a more effective method to increase the number of surface reactive sites and visible light absorption of graphitic carbon nitride through defect control.

In this work, we designed a new structure with defect engineering modifying atomic-layered g-C_3N_4 to coordinate the game relationship between reaction sites and visible light absorption of atomic-layered g-C_3N_4. The extended light absorption increased the number of reaction sites and shortened the migration distance of photogenerated electrons/holes which suppressed recombination of charge carriers and ultimately led to greatly improved photocatalytic hydrogen evolution activities under visible light. Thus, the defect-modified atomic-layered g-C_3N_4 exhibited much higher H_2 evolution activity (3.7 $mmol \cdot g^{-1} \cdot h^{-1}$) under visible-light irradiation compared with pristine g-C_3N_4. In addition, this method would inspire more similar attempts for atomic-layered 2D nanomaterial modification.

2. EXPERIMENTAL SECTION

2.1. Sample Preparation. *2.1.1. Preparation of Bulk g-C_3N_4.* Bulk g-C_3N_4 was synthesized by direct thermal polymerization of dicyandiamide in a chemical vapor deposition furnace.[2] Typically, 5 g of dicyandiamide in sealed ceramic crucibles was calcined at 550 °C for 4 h with 2 °C/min heating rate under an Ar atmosphere. The brilliant yellow product was naturally cooled down to ambient temperature and ground into powder for later use. The product is denoted as B-g-C_3N_4.

2.1.2. Preparation of Atomic-Layered g-C_3N_4. The samples of atomic-layered g-C_3N_4 were synthesized by the two-step thermal exfoliation method. Typically, 500 mg of B-g-C_3N_4 was first put into the ceramic crucibles and heated at 550 °C for 1 h with a ramp rate 5 °C/min. After natural cooling, the samples were further heated at 500 °C for 2 h with a ramp rate of 2 °C/min. Finally, the sample was collected and denoted as A-g-C_3N_4.

2.1.3. Preparation of Defect Engineering-Modified Atomic-Layered g-C_3N_4. Defect engineering-modified atomic-layered g-C_3N_4 was synthesized using high-temperature treatment and the two-step thermal exfoliation process. Typically, 5 g of dicyandiamide in sealed ceramic crucibles was calcined at 650 °C for 4 h with 2 °C/min heating rate under an Ar atmosphere. The obtained orange red product was ground into powder. Then, 500 mg of the product was put into the ceramic crucibles and heated at 550 °C for 1 h with a ramp rate of 5 °C/min. After being naturally cooled, the samples were further heated at 500 °C for 2 h with a ramp rate 2 °C/min. Finally, the sample was collected and denoted as A-V-g-C_3N_4.

2.2. Photocatalytic Activity Experiments. The photocatalytic H_2 evolution experiments were tested in a 250 mL Pyrex cell. The cell was connected to a closed gas evacuation and circulation system. The PLS-SXE300CBF Xe arc lamp (Beijing Perfectlight Co. Ltd.) was used as a light source with a 420 nm cut-off filter.

In a typical photocatalytic experiment, 50 mg of photocatalyst powders were suspended in 100 mL of an aqueous solution containing 10% triethanolamine as a scavenger. An in situ photodeposition method was introduced to load Pt as the cocatalyst. H_2PtCl_6 was used as the Pt source, and 132 μL H_2PtCl_6 (1% wt Pt) was dissolved into the solution directly before the evacuation process started. The mixed solution was then irradiated under stirring for 60 min fully to deposit Pt nanoparticles on the photocatalyst. The solution was evacuated, and generated gas was collected under irradiation of a Xe lamp with a cooling system to maintain its constant temperature. The gas was tested by gas chromatography (Shimadzu, GC-2014, nitrogen carrier). The apparent quantum yield (AQY) was calculated using the following eq 1.

$$\text{AQY (\%)} = \frac{2 \times n(H_2)}{n(\text{photons})} \times 100\% \tag{1}$$

where $n(H_2)$ and n(photons) denote the number of produced H_2 molecules and the number of incident photons, respectively. According to the previous report, n(photons) was 2.9×10^{20} photon h^{-1}.[28]

2.3. Characterizations. The structure and morphology of all samples were characterized using a X-ray diffraction (XRD) diffractometer (DX-2700, Dandong, China), Fourier transform infrared spectroscopy (FTIR, model Magna 560, Thermo Fisher Scientific, USA), transmission electron microscopy (TEM, FEI Tecnai G2 F20), and atomic force microscopy (AFM). The nitrogen vacancies were measured by energy-dispersive X-ray spectroscopy (EDS) , X-ray photoelectron spectroscopy (XPS, ESCALAB 250Xi, Al Kα, 1486.6 eV, Thermo Fisher Scientific, USA), and the content of hydroxyl radicals ($^{\bullet}$OH) and superoxide anion radicals ($^{\bullet}O_2^-$) were detected by electron spin resonance (ESR, JEOL JES-FA200) with 5,5-dimethyl-L-pyrroline *N*-oxide (DMPO) as the active-species capture agent under the condition of visible light irradiation for 10 min. The specific surface area was determined by nitrogen adsorption/desorption measurements with an automated surface area (QUADRASORB SI, Quantachrome, America). The optical and photoelectric properties were determined using a UV−vis diffuse reflection spectrophotometer (190−900 nm) and photoluminescence (PL) spectra (FLS1000 Edinburgh fluorescence spectrometer, excitation wavelength at 325 nm).

3. RESULTS AND DISCUSSION

3.1. Structural and Morphological Characteristics. XRD was used to analyze the crystallographic structure of the A-V-g-C_3N_4 and A-g-C_3N_4 catalyst, as shown in Figure 1. It can be seen from Figure 1 that the peak at 27.5° belongs to the (002) plane reflections of the interlayer stacking,[24,29,30] while the peak at 12.8° should be attributed to the (100) plane reflections of the in-plane repeated units and is very weak. This result indicates that the ordered structures restack along the

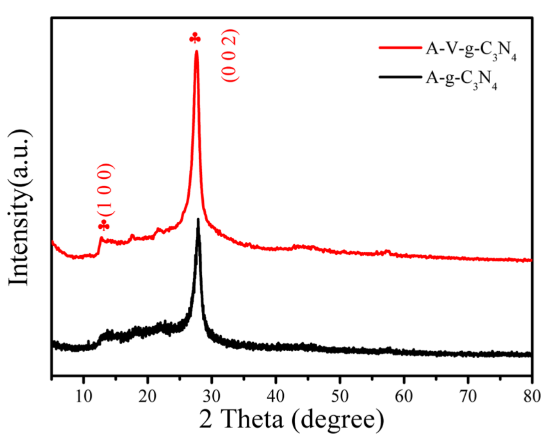

Figure 1. XRD pattern of A-V-g-C_3N_4 and A-g-C_3N_4 samples.

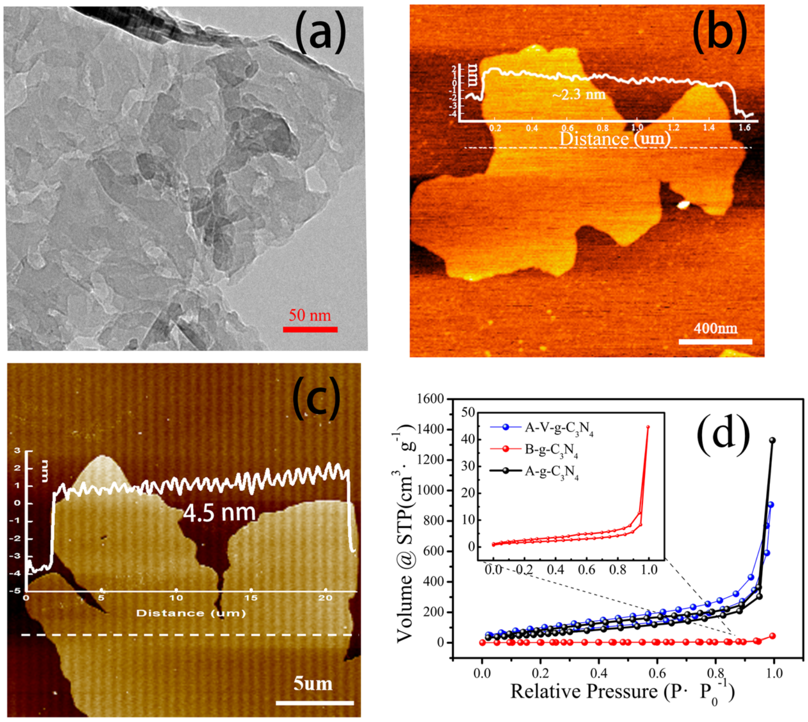

Figure 2. TEM images of A-V-g-C_3N_4 (a), AFM images of A-V-g-C_3N_4 (b), A-g-C_3N_4 (c), and N_2 physisorption isotherms (d).

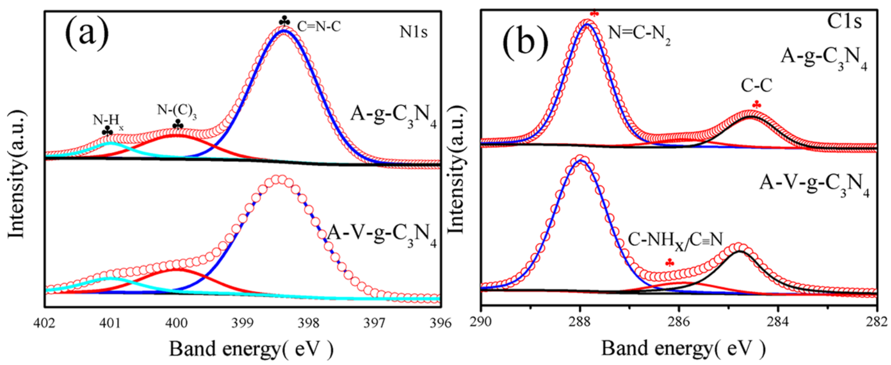

Figure 3. High-resolution XPS spectra of N 1s (a), C 1s (b) of A-g-C_3N_4 and A-V-g-C_3N_4 samples.

(001) direction of the A-V-g-C_3N_4 and A-g-C_3N_4 nanosheet.[29,31]

The morphology and thickness of the A-V-g-C_3N_4 nanosheets were investigated by TEM and AFM, as shown in Figure 2. The TEM images in Figure 2a show that the nanosheets of the A-V-g-C_3N_4 sample were obtained. This is probably a consequence that the bulk g-C_3N_4 was exfoliated into small nanosheets in the two-step thermal exfoliation process. Figure 2b,c show the AFM images of A-V-g-C_3N_4 and A-g-C_3N_4 nanosheets, respectively. The representative thicknesses of A-V-g-C_3N_4 and A-g-C_3N_4 samples are 2.3 and 4.5 nm, respectively. The results show that the atomic-layered g-C_3N_4 photocatalysts can be prepared by two-step thermal exfoliation methods. In addition, the migration distance of photogenerated electrons/holes was dramatically shortened, in which more electrons participate during hydrogen evolution reaction reaction.

The N_2 sorption isotherms of B-g-C_3N_4, A-g-C_3N_4, and A-V-g-C_3N_4 samples showed a typical curve with the hysteresis loop of the type IV isotherms in Figure 2d.[32] The Brunauer−Emmett−Teller (BET) surface areas of all samples were calculated, as shown in the Supporting Information (Table S1). The BET surface area increases from 6.1 $m^2 \cdot g^{-1}$ of B-g-C_3N_4 to 223 $m^2 \cdot g^{-1}$ of A-g-C_3N_4 and 267.1 $m^2 \cdot g^{-1}$ of A-V-g-C_3N_4. According to the result of the N_2 sorption isotherms, we can speculate that the sample of A-V-g-C_3N_4 may show superior photocatalytic activity because the large surface area could expose more active sites and increase solid−liquid interfaces that enhance the transportation of photocatalyst-relevant species.[33,34]

The existence of defect engineering on the sample of A-V-g-C_3N_4 was confirmed by the results of EDS, XPS, and ESR. The results of elemental composition for A-V-g-C_3N_4 by the EDS method are listed in the Supporting Information (Table S1). It can be found that the weight ratio of C/N increases from 0.67 of B-g-C_3N_4 to 0.83 of A-V-g-C_3N_4. This result suggests that the N vacancies may exist with A-V-g-C_3N_4. In order to confirm that the N vacancies exist with A-V-g-C_3N_4, XPS analysis was conducted on the samples of A-g-C_3N_4 and A-V-g-C_3N_4. Figure 3a,b shows the high-resolution XPS spectra results of all the samples. The N 1s XPS results of A-g-C_3N_4 and A-V-g-C_3N_4 are shown in Figure 3a. The peaks of 398.7,

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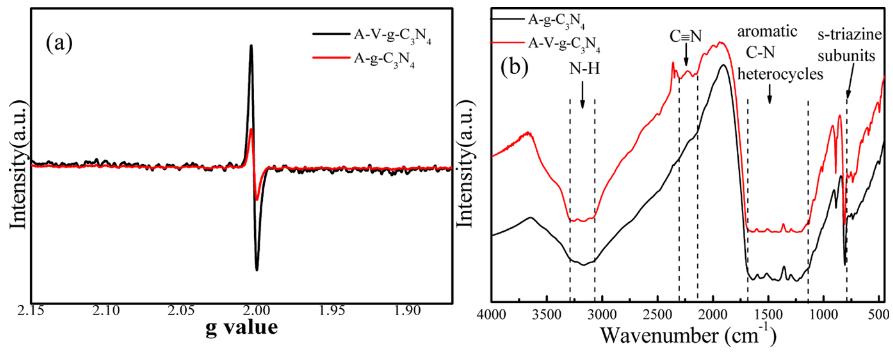

Figure 4. ESR signals of N vacancy (a), FTIR spectra of the A-g-C_3N_4 and A-V-g-C_3N_4 samples (b).

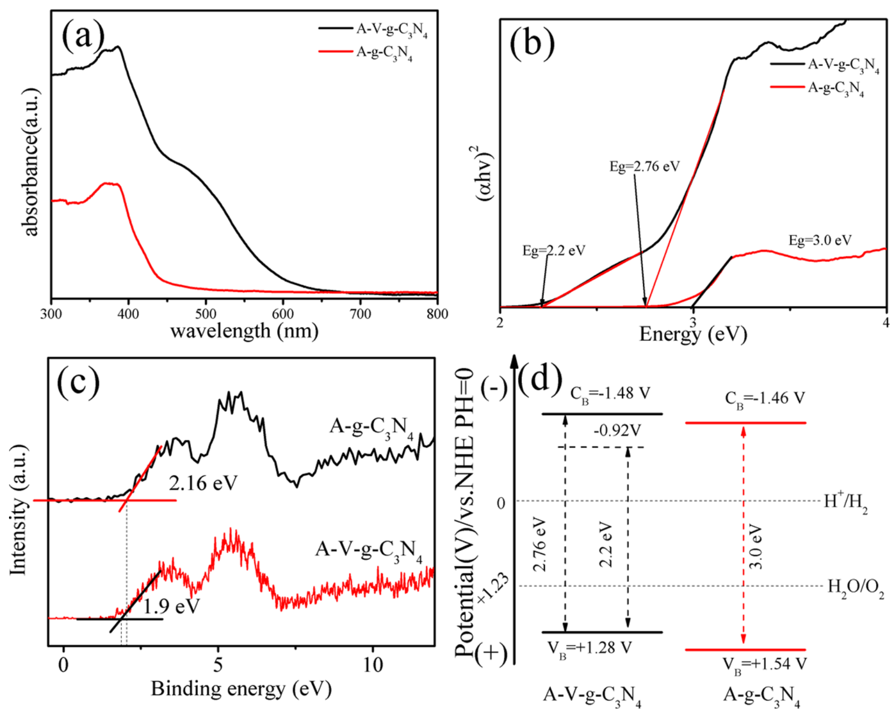

Figure 5. UV−vis absorption spectra (a), band gap energies (b), valence band XPS spectra (c), and band structure alignments of the A-g-C_3N_4 and A-V-g-C_3N_4 samples (d).

400, and 401 eV should be attributed to the sp^2-hybridized nitrogen (N_{2c}), sp^3-tertiary nitrogen (N_{3c}), and NH_x groups in the heptazine framework.[35−38] The area ratio of the (N_{2c})/(N_{3c}) was calculated and is shown in the Supporting Information (Table S1). It shows that ratio decreases from 3.3 of A-g-C_3N_4 to 1.84 of A-V-g-C_3N_4. This result strongly demonstrates that the N vacancies were formed in N_{2c} sites, which is consistent with the literature studies.[35,36] The C 1s XPS results of A-g-C_3N_4 and A-V-g-C_3N_4 are shown in Figure 3b. It can be found that the peaks located at 288.0, 285.9, and 284.7 eV are attributed to N═C−N_2 coordination in the framework of g-C_3N_4, C-NH_x, or cyanogen groups on the edges of heptazine units and typically assigned to C−C bonds, respectively.[27,39] Compared to A-g-C_3N_4, the intensity of the peak at 285.9 eV in A-V-g-C_3N_4 is significantly enhanced. This can be taken as an additional evidence for the formation of cyanogen groups because C≡N groups possess similar C 1s binding energies to C−NH_x.[40] In addition, the behaviors of nitrogen vacancies of A-g-C_3N_4 and A-V-g-C_3N_4 samples were evaluated by the ESR, as shown in Figure 4a. Compared with A-g-C_3N_4, the ESR spectrum of A-V-g-C_3N_4 displays a stronger symmetrical signal cantering at a g factor of 2.002 (g = 2.002), indicating the generation of unpaired electrons on the carbon atoms of the aromatic ring within π-bonded nanosized clusters and nitrogen vacancies. This result further demonstrates the presence of the nitrogen vacancies in A-V-g-C_3N_4.[41,42] Note that an analogous signal is also detected for A-g-C_3N_4. It is possible to generate partial disorders (nitrogen defects) of A-g-C_3N_4 in the thermal exfoliation process, which is in accordance with the literature.[43,44]

In addition to N defects, cyanogen groups (C≡N) can be formed under the high-temperature treatment process. This could be well proved by the FTIR spectrum, as shown in Figure 4b. The typical characteristic peaks of g-C_3N_4 can be observed in all samples. The peaks at 3000−3500, 1000−1700, and 810 cm^{-1} are attributed to the vibrational absorptions of N−H bonds, aromatic C−N heterocycles, and *s*-triazine subunits, respectively.[45−47] Compared with the spectrum of A-g-C_3N_4, a new peak appeared in 2160−2190 cm^{-1} on the spectrum of the A-V-g-C_3N_4 sample. The peak belongs to stretching vibrations of the cyanogen groups (C≡N).[25,46] This result indicates that the cyanogen groups (C≡N) were formed

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by the high-temperature treatment. Combining with comprehensive analysis of EDS, XPS, ESR, and FTIR, the defect engineering of nitrogen vacancies and cyanogen groups in A-V-g-C_3N_4 is validated.

3.2. Photophysical and Photochemical Properties. The optical and photoelectric properties of A-g-C_3N_4 and A-V-g-C_3N_4 samples were investigated by UV−vis diffuse reflectance spectra (DRS), PL, and valence band (VB) XPS. The results of the UV−vis spectra of A-g-C_3N_4 and A-V-g-C_3N_4 samples are shown in Figure 5a. Compared with the DRS result of A-g-C_3N_4, A-V-g-C_3N_4 shows a redshift of the absorption edge and stronger light absorption below the wavelength of 600 nm. In addition, the band gaps (E_g) of all the samples were calculated using the Kubelka−Munk function, as shown in Figure 5b and Table S1. Compared with the band gap of g-C_3N_4, the band gap of A-g-C_3N_4 was enlarged to 3.01 eV. This is due to the quantum size effect.[48−50] However, when the nitrogen vacancies and cyanogen groups were introduced into A-g-C_3N_4, the band gap was decreased to 2.76 eV for A-V-g-C_3N_4. This result suggests that the quantum size effect is restrained. In addition, the A-V-g-C_3N_4 sample shows a very strong tail absorption (Urbach tail) in the visible-light region. As we all know, the Urbach tail is attributed to the electronic states located within the band gap (known as mid-gap states).[51−53] Furthermore, the mid-gap state energy (marked as E_g') for A-V-g-C_3N_4 is 2.2 eV, which was calculated by the Kubelka−Munk method. According to the previous works, the mid-gap states at 0.5−1.0 eV below the conduction band minimum are created, when the cyanogen groups were introduced to g-C_3N_4.[54,55] Therefore, E_g' is located at 2.2 eV above the VB position. Hence, the empty mid-gap states can accommodate the electrons photo-excited from the VB of A-V-g-C_3N_4, which greatly contribute to the absorption of photons with energies smaller than the band gap, thus resulting in the appearance of the Urbach tail.

The band structures of the A-g-C_3N_4 and A-V-g-C_3N_4 samples were examined by VB XPS spectra analysis, as shown in Figure 5c. The VB XPS result suggests that the difference energy (ΔE) between the Fermi level (E_F) and VB maximum (V_B) are 2.16 and 1.9 eV for A-g-C_3N_4 and A-V-g-C_3N_4, respectively. In addition, the VB potential (E_{VB}) and conduction band potential (E_{CB}) of A-g-C_3N_4 and A-V-g-C_3N_4 were calculated by the formula $E_{VB} = \Delta E + \Phi - E_{vac}$ and $E_{CB} = E_{VB} - E_g$, where E_{vac} is the energy of free electrons in the hydrogen scale with a constant of 4.5 eV, E_g is the band gap of samples, and Φ is the work function. The work function, Φ, is 3.88 eV for g-C_3N_4.[2,30,56] Hence, the V_B and C_B positions of A-g-C_3N_4 were determined as 1.54 and −1.46 eV, respectively. The V_B, mid-gap state, and C_B positions of A-V-g-C_3N_4 were calculated as 1.28, −0.92, and −1.48 eV, respectively. These results of band structures are summarized in Table S1 and Figure 5d. The results of VB XPS spectra indicate that the nitrogen defects and cyanogen groups can up-shift the VB and form the mid-gap state energy level, respectively.

The separation efficiency of photogeneration electron/hole pairs for the A-g-C_3N_4 and A-V-g-C_3N_4 samples were investigated by PL spectroscopy. From Figure 6, the emission peak at 435 nm of the A-g-C_3N_4 was attributed to the direct electron and hole recombination of band transition. Compared with A-g-C_3N_4, the emission peak in the sample of A-V-g-C_3N_4 became weaker and red-shifted to 452 nm. The results indicate that the recombination of photogenerated electrons and holes in A-V-g-C_3N_4 was greatly inhibited.

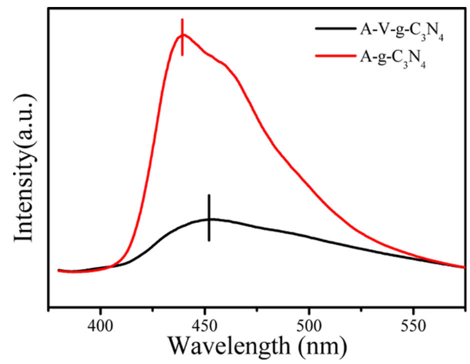

Figure 6. PL emission spectra of the A-V-g-C_3N_4 and A-g-C_3N_4 samples.

It is speculated that the redox activity of A-V-g-C_3N_4 is better than that of A-g-C_3N_4 under visible light. The inference was demonstrated by ESR, as shown in Figure 7a,b. The ESR signals with the intensity ratio of 1:2:2:1 in Figure 7a can be considered as the feature of DMPO−$^{\bullet}$OH, and the intensity ratio of 1:1:1:1 in Figure 7b is classified as the characteristics of DMPO−$^{\bullet}O_2^-$. The $^{\bullet}$OH and $^{\bullet}O_2^-$ radical signal for the A-V-g-C_3N_4 are stronger than A-g-C_3N_4. These results demonstrate that the A-V-g-C_3N_4 sample has superior oxidation and reduction activity for the photocatalytic reactions.

3.3. Photocatalytic H_2 Production Activity. Figure 8a shows the photocatalytic H_2 production activities of all samples. It can be found that the H_2 evolution rate and the AQY of B-g-C_3N_4 are 0.284 mmol·g^{-1}·h^{-1} and 1.15% ($\lambda > 420$ nm), respectively. It is due to the low surface area and poor charge transport capability. When the B-g-C_3N_4 samples were exfoliated to A-g-C_3N_4, the H_2 evolution rate and the AQY obviously increases to 2.69 mmol·g^{-1}·h^{-1} and 10.89% ($\lambda > 420$ nm), respectively. Although the light-harvesting ability of A-g-C_3N_4 was inhibited, the surface areas were enlarged, and the electron/hole transfer distances were shortened. Hence, the H_2 evolution rate was enhanced. In addition, the H_2 evolution rate and AQY of A-V-g-C_3N_4 obtained under optimized preparation conditions reached up to 3.7 mmol·g^{-1}·h^{-1} and 14.98% ($\lambda > 420$ nm), respectively. It is higher than other's work listed in the Supporting Information (Table S2). In addition, the stability of the A-V-g-C_3N_4 photocatalyst is evidenced by the long-term H_2 evolution; as shown in Figure 8b, only a slight decrease was observed during the 24 h test. This result shows that the catalyst of A-V-g-C_3N_4 has excellent photochemical stability. The results of DRS, PL, and ESR indicated that the narrowed band gap and mid-gap states effectively enhanced visible-light harvesting and decreased recombination of electrons and holes. Hence, the H_2 evolution rate of A-V-g-C_3N_4 was remarkably enhanced.

4. CONCLUSIONS

In summary, the defect engineering modifying atomic-layered g-C_3N_4 nanosheets (A-V-g-C_3N_4) were successfully synthesized via a high-temperature treatment and two-step thermal exfoliation methods. Detailed analysis shows that the defect engineering of nitrogen vacancies and cyanogen groups was introduced into the g-C_3N_4 structure. The defects can cause the upshift of the VB potential and form mid-gap states which effectively suppress the quantum size effect of atomic-layered g-C_3N_4 that increase visible light response and decrease recombination of electrons and holes. In addition, the ultrathin

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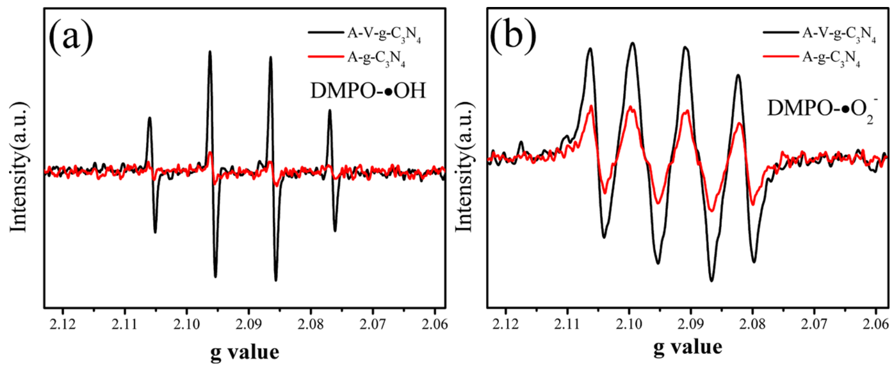

Figure 7. ESR signals of DMPO−•OH (a) and DMPO−•O_2—(b) under visible light irradiation of the A-g-C_3N_4 and A-V-g-C_3N_4 samples.

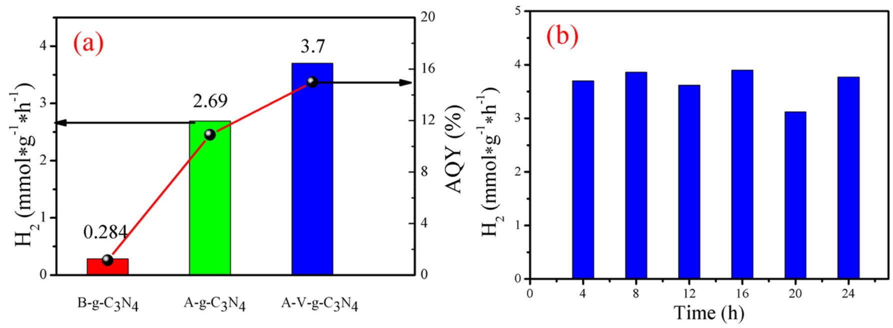

Figure 8. H_2 evolution rates of the B-g-C_3N_4, A-g-C_3N_4, and A-V-g-C_3N_4 samples (a) and cycle test of H_2 evolution over A-V-g-C_3N_4 (b).

thickness (2.3 nm) significantly increases the surface areas (267.1 $m^2 \cdot g^{-1}$) of g-C_3N_4 which provided more active sites for photocatalytic reaction and decreases the electron/hole transfer distances. Consequently, the photocatalytic activity was remarkably enhanced. This work put forward a promising strategy to solve the drawback of ultrathin g-C_3N_4 nanosheets, such as the quantum size effect, and maintain its advantage, for example, large surface areas. In addition, it would inspire more similar attempts for atomic-layered 2D nanomaterial modification.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.9b21115.

Elemental composition, BET surface area, XPS spectra and valence band XPS spectra results of B-g-C_3N_4, A-g-C_3N_4, and A-V-g-C_3N_4 sample. Comparison of H_2 evolution rate of A-V-g-C_3N_4 and other g-C_3N_4 photocatalysts that are recently reported (PDF)

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Notes

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